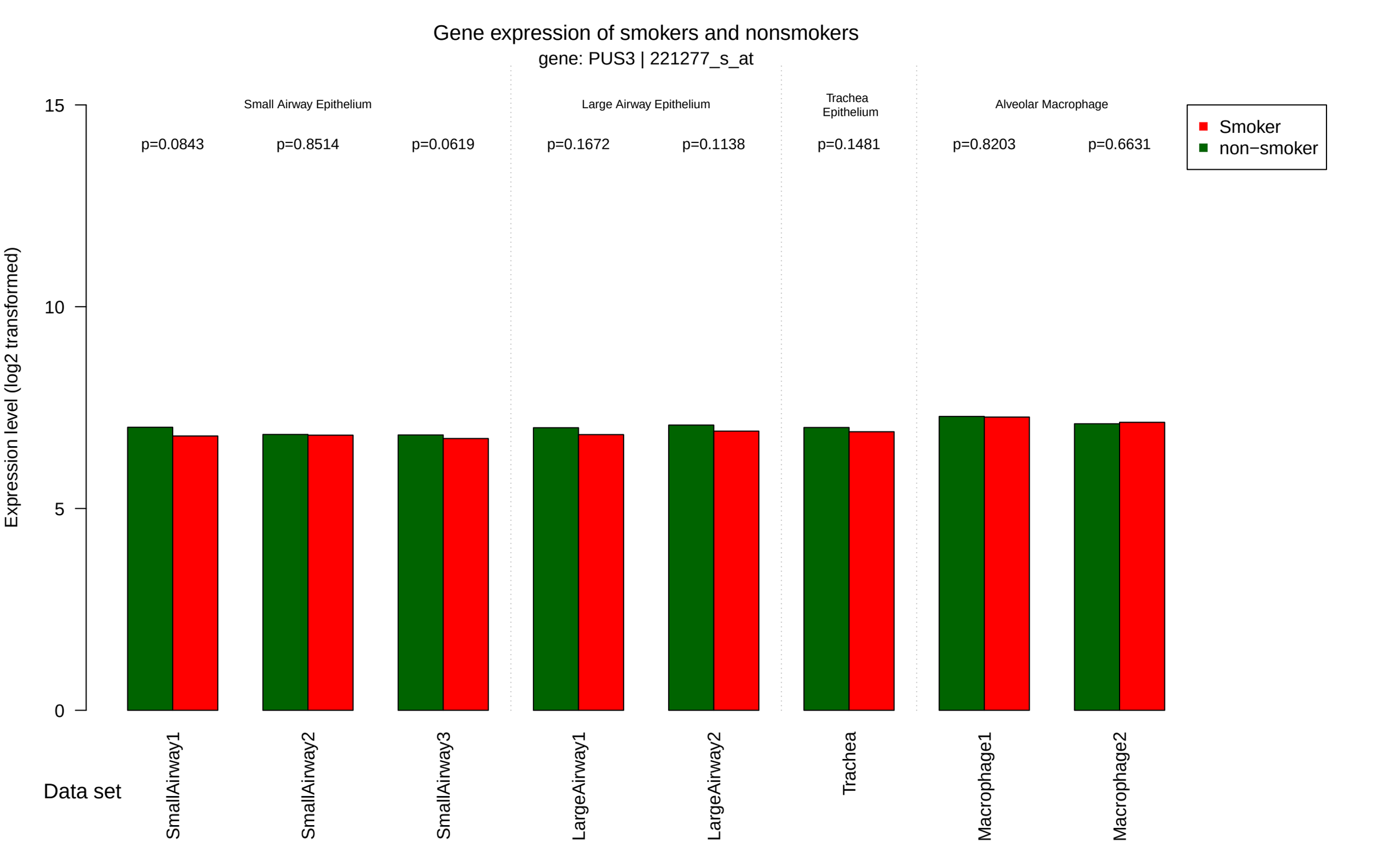
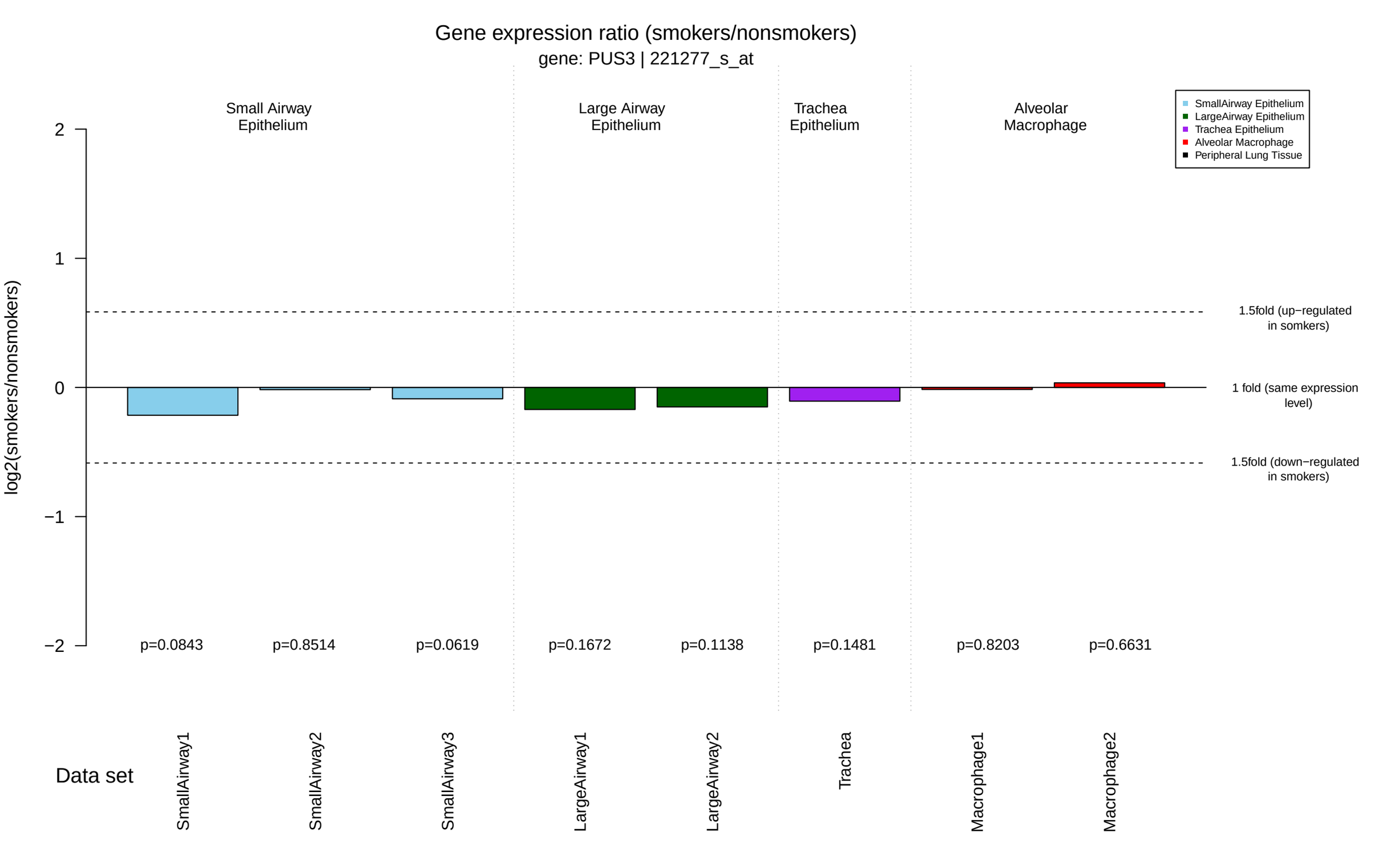
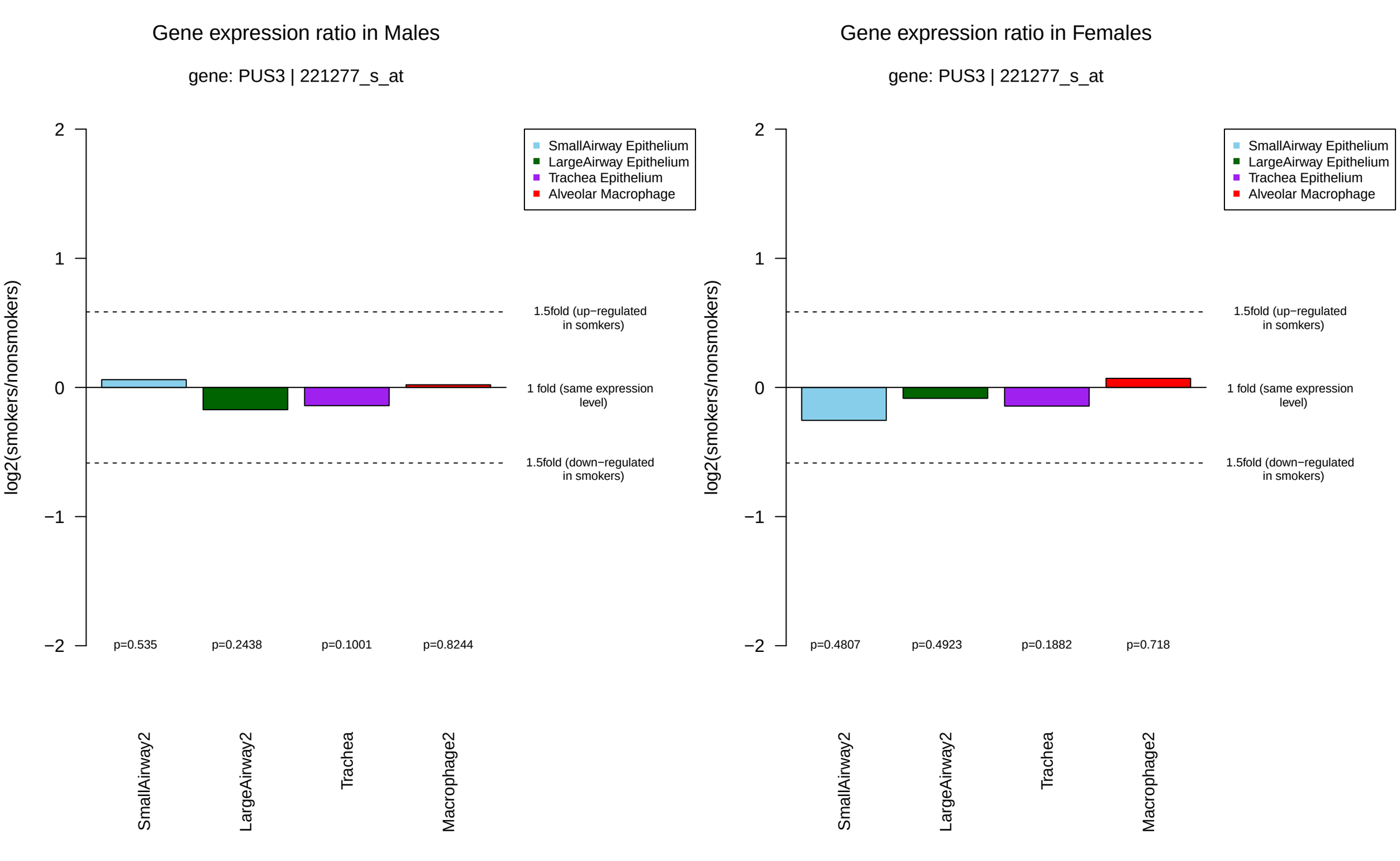
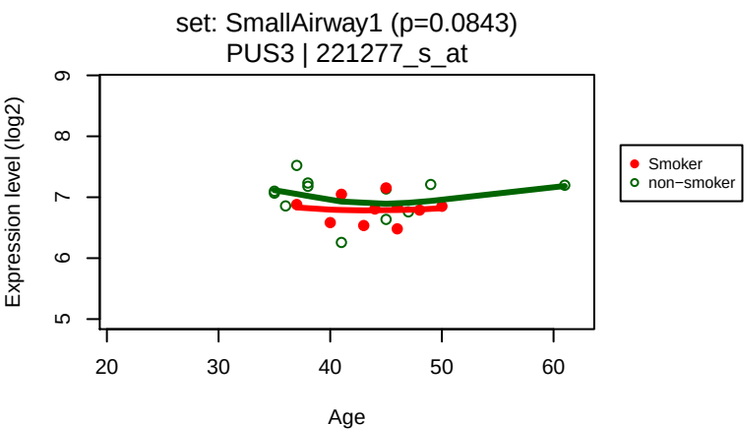
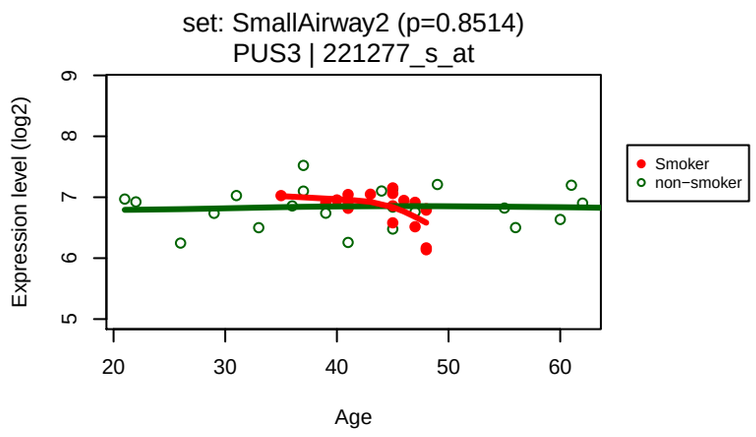
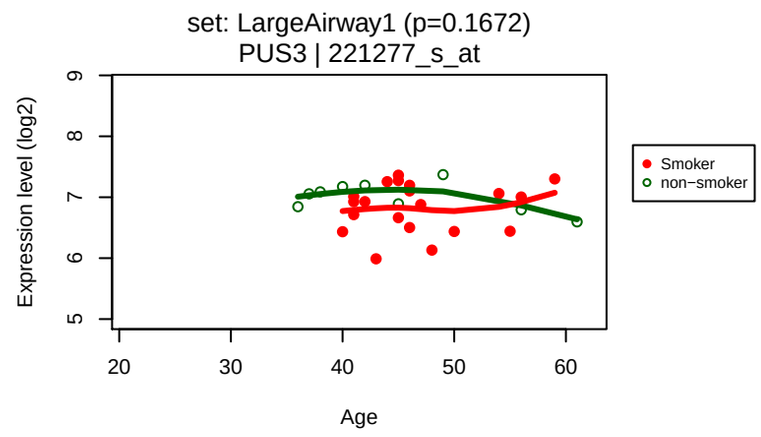

Gene expression of smokers and nonsmokers
gene: PUS3 | 221277_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0843
p=0.8514
p=0.0619
p=0.1672
p=0.1138
p=0.1481
p=0.8203
p=0.6631
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

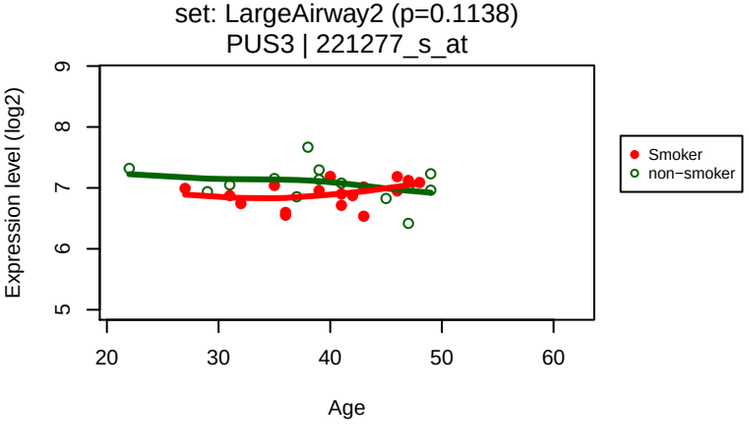

Gene expression ratio (smokers/nonsmokers)
gene: PUS3 | 221277_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0843
p=0.8514
p=0.0619
p=0.1672
p=0.1138
p=0.1481
p=0.8203
p=0.6631
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

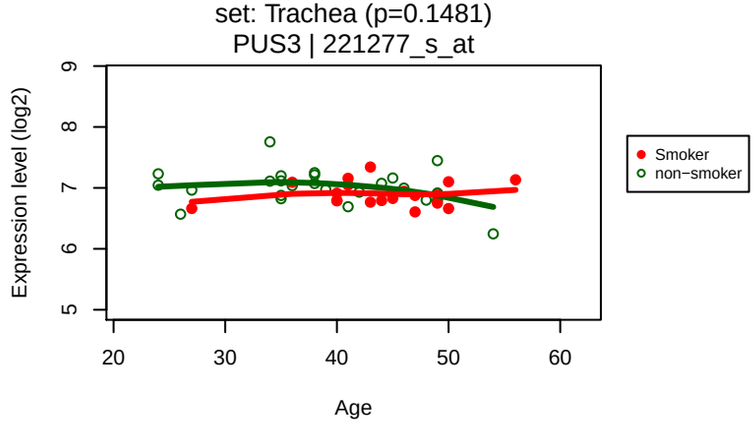

Gene expression ratio in Males
gene: PUS3 | 221277_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.535
p=0.2438
p=0.1001
p=0.8244
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PUS3 | 221277_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4807
p=0.4923
p=0.1882
p=0.718
SmallAirway2
LargeAirway2
Trachea
Macrophage2

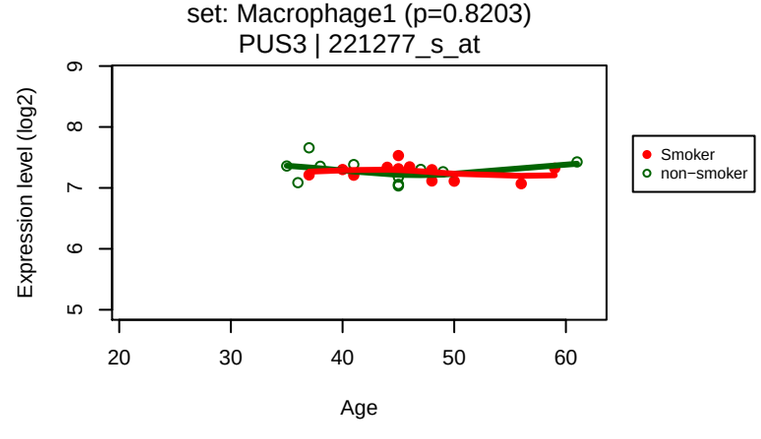

set: SmallAirway1 (p=0.0843)
PUS3 | 221277_s_at
Expression level (log2)
Age
Smoker
non−smoker

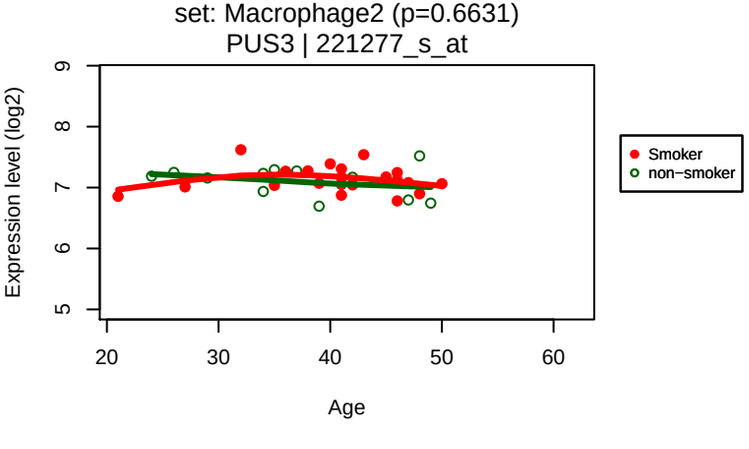

set: SmallAirway2 (p=0.8514)
PUS3 | 221277_s_at
Expression level (log2)
Age
Smoker
non−smoker

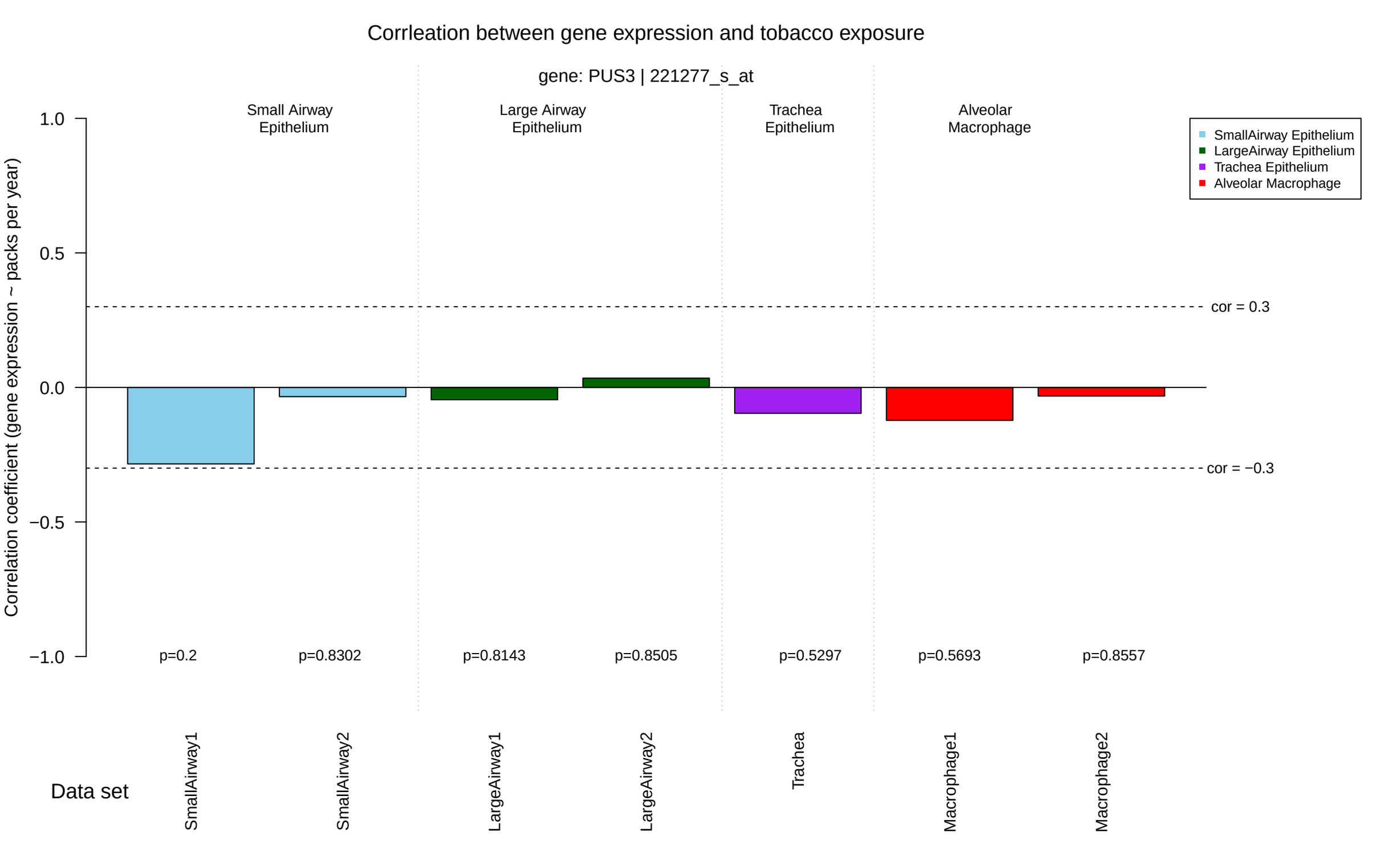

set: LargeAirway1 (p=0.1672)
PUS3 | 221277_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1138)
PUS3 | 221277_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1481)
PUS3 | 221277_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8203)
PUS3 | 221277_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6631)
PUS3 | 221277_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PUS3 | 221277_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2
p=0.8302
p=0.8143
p=0.8505
p=0.5297
p=0.5693
p=0.8557
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2